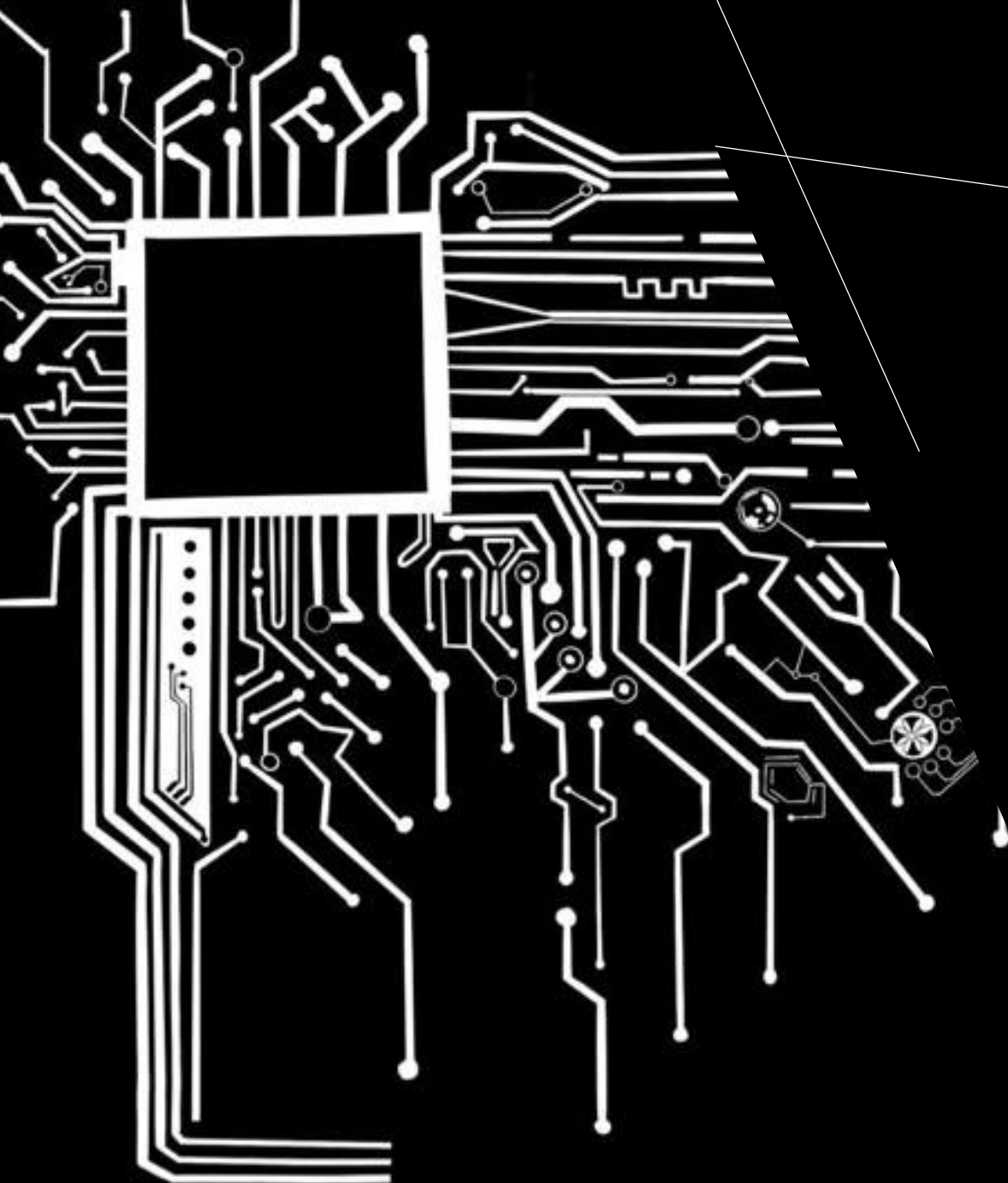


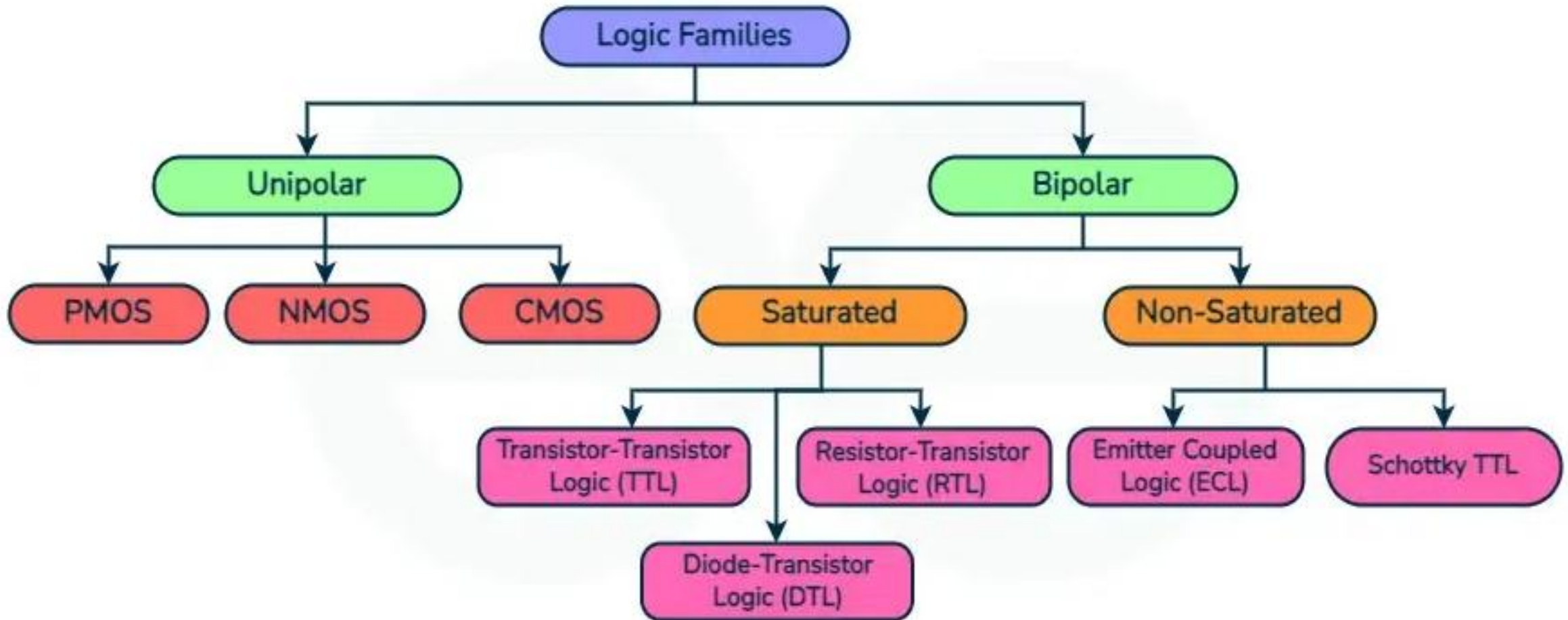


ICE 2343

DIGITAL LOGIC DESIGN

ABHIRUP SARKER
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DEPARTMENT OF ROBOTICS AND MECHATRONICS
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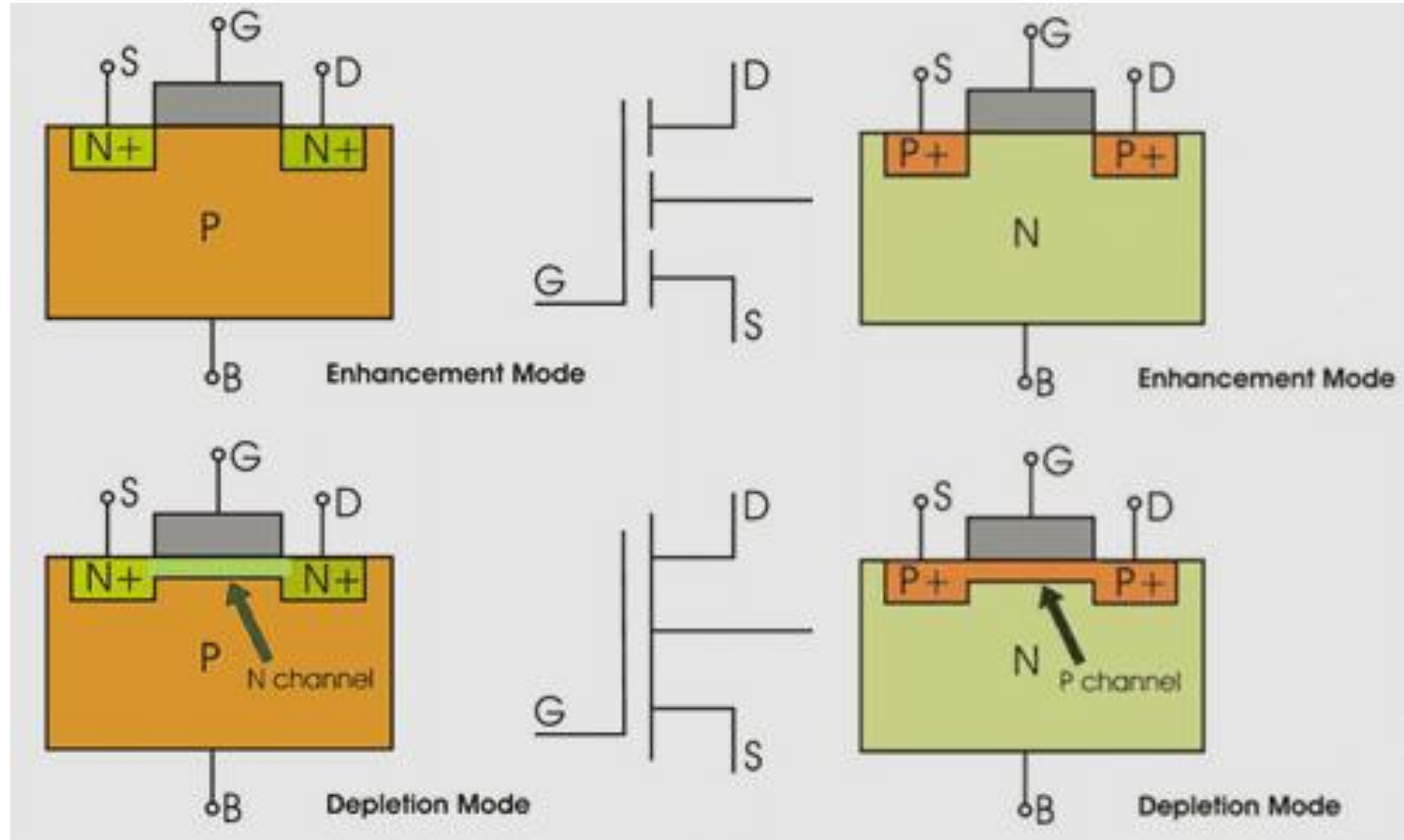
Digital Logic Families



Characteristics

1. **Fan IN:** maximum no. of inputs that can be applied to the logic gate for proper operation without error or failure.
2. **Fan OUT:** no. of loads that output gate can drive without failing or error.
3. **Speed:** time between application of input and change in output.
4. **Power dissipation:** power consumed or dissipated per gate during processing or operation.
5. **Propagation delay:** the average delay time for signal to propagate input to output.
6. **Noise margin:** it is the limit of noise voltage that can be present without damaging proper operation of circuit. Shows how well noise can be eliminated.

MOSFET

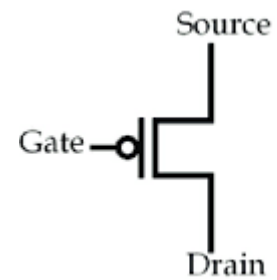
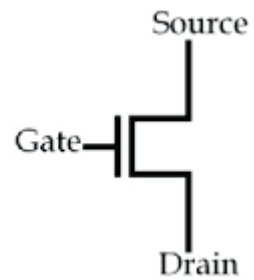


NMOS & PMOS

NMOS

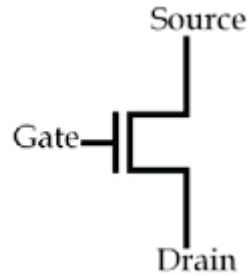


PMOS



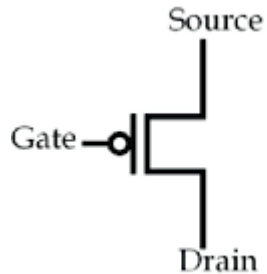
Symbols

NMOS & PMOS



NMOS

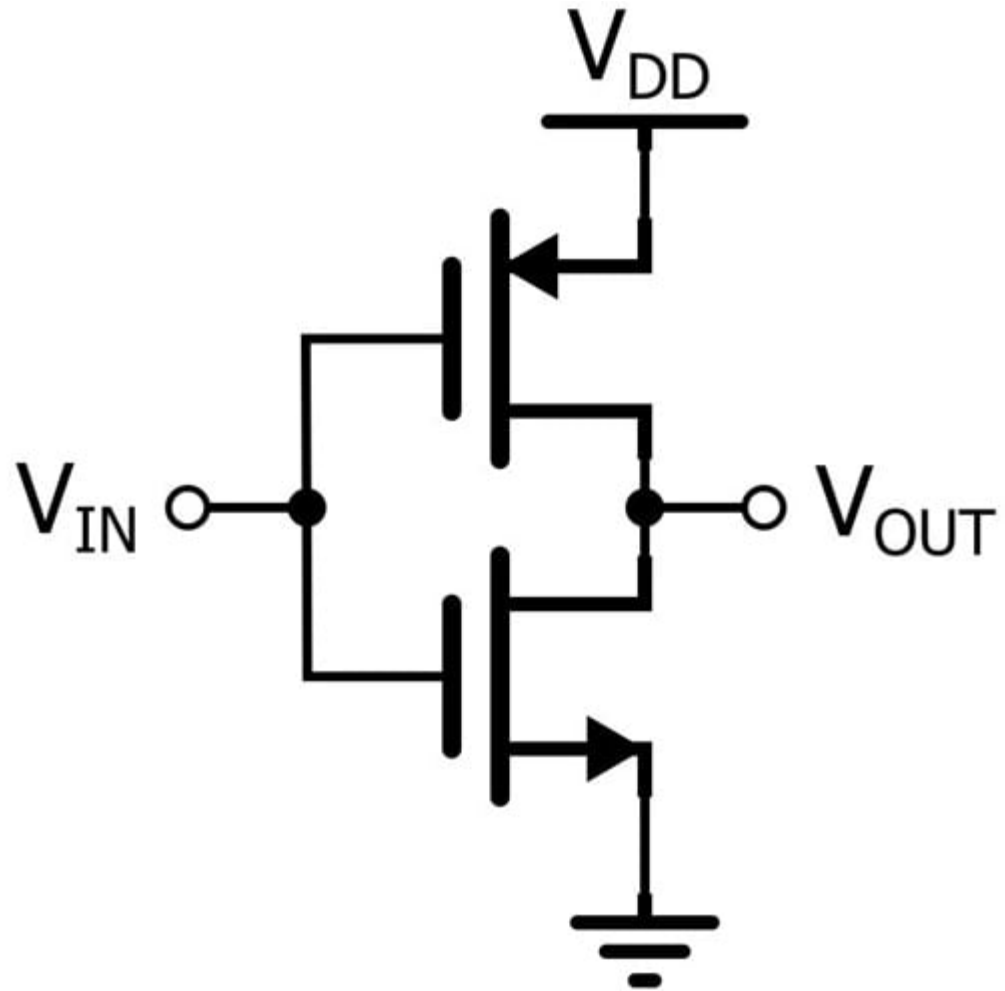
If input is 0(low), drain to source open circuit i.e. current will not flow
If input is 1(high), drain to source short circuit i.e. current will flow



PMOS

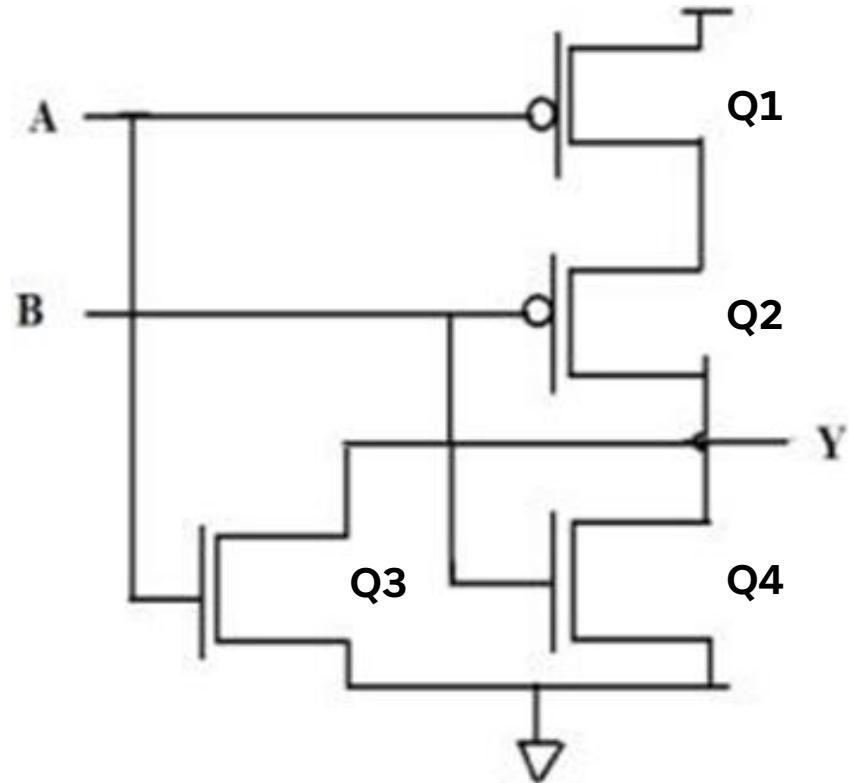
If input is 0(low), source to drain short circuit i.e. current will flow
If input is 1(high), source to drain open circuit i.e. current will not flow

NOT Gate



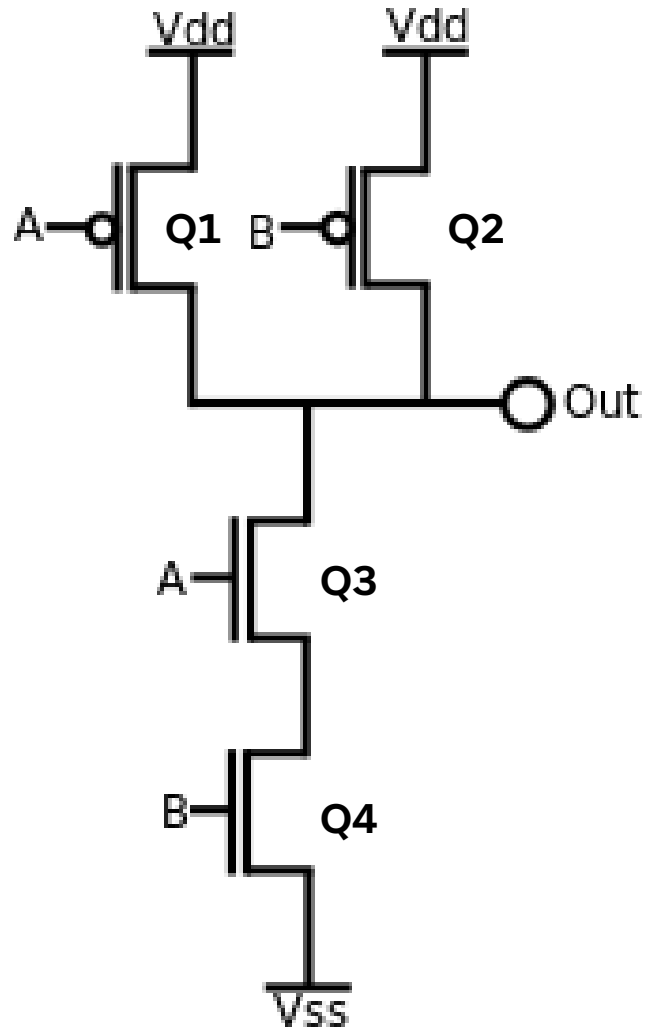
V_{in}	PMOS	NMOS	V_{out}
0	SC	OC	1
1	OC	SC	0

NOR Gate



A	B	Q1	Q2	Q3	Q4	Y
0	0	on	on	off	off	1
0	1	on	off	off	on	0
1	0	off	on	on	off	0
1	1	off	off	on	on	0

NAND Gate



A	B	Q1	Q2	Q3	Q4	Y
0	0	on	on	off	off	1
0	1	on	off	off	on	1
1	0	off	on	on	off	1
1	1	off	off	on	on	0

Practice

1. Draw the AND & OR gate circuits using NMOS & PMOS.
2. Explain how NOR gate circuit works for input $A=0$, $B=0$.

Reference

Digital Systems Principles and Applications by Ronald J.Tocci
Follow your class lecture notes

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract shape on the left side of the slide.

THANK YOU